GATTU RISHI PRIYESH

AI/ML Engineer | RAG Systems, Agentic AI and LLM Applications | Prompt Engineering | Generative AI

+91 7780643855 | rishipriyesh5603@gmail.com

LinkedIn | GitHub | HackerRank

PROFESSIONAL SUMMARY

Generative AI and ML Engineer with a B.Tech in Computer Science, specialising in RAG pipelines, LangChain-based LLM applications and vector database systems. Experienced in developing end-to-end AI solutions involving **LLMs, embeddings, vector databases, document processing, REST APIs, Streamlit, FastAPI, Docker, and cloud deployment**. Strong foundation in machine learning algorithms, CNNs, data preprocessing, model evaluation, and AI application development. Interested in building production-oriented AI systems and solving real-world problems using modern AI technologies. Sanfoundry Python certified. Targeting AI/ML Engineer, Data Scientist and GCP Cloud Engineer roles.

TECHNICAL SKILLS

GenAI and Agentic AI	LangChain, LLaMA 3, RAG Pipelines, Agentic AI, Prompt Engineering, Hugging Face Transformers, FAISS, ChromaDB, Vector Databases, OLLAMA
ML and Deep Learning	Scikit-learn, Supervised and Unsupervised Learning, TensorFlow, Keras, Neural Networks, Image Classification, TFIDF, NLP, ANN,CNN, LSTM, PyTorch
Deployment and APIs	FastAPI, Docker, GCP (Vertex AI, Cloud Storage), REST API Design, Vercel, Render
Data Analysis	Pandas, NumPy, Matplotlib, Seaborn, EDA, Power BI, Tableau
Languages	Python (Intermediate), SQL (Intermediate), JavaScript (Intermediate)
Databases	MySQL, PostgreSQL
Tools	Git, GitHub, Jupyter Notebook, VS Code, Google Colab

PROFESSIONAL EXPERIENCE

Generative AI Intern, Datavalley Nov 2025 - Present

- Built a document question-answering system using LangChain and LLaMA 3, integrating FAISS vector database for semantic document retrieval - enabling accurate context-grounded answers from enterprise documents
- Developed and optimised a RAG pipeline performing document chunking, embedding generation using Hugging Face embeddings, and similarity search with FAISS and ChromaDB for high-dimensional vector retrieval
- Developed REST APIs using FastAPI to expose AI model inference endpoints for sentiment analysis and document Q&A applications, enabling frontend and third-party integration
- Built interactive Gradio demo interfaces allowing stakeholders to upload documents and query the RAG pipeline live through a browser interface
- Applied prompt engineering techniques to improve LLM response quality and reduce hallucination rates across multiple use cases
- Deployed AI applications using FastAPI with Docker containerisation on Google Cloud Platform, managing cloud storage and service configuration

KEY PROJECTS

1. AI Knowledge Assistant — RAG-Based Document Intelligence Platform

Python | FastAPI | Streamlit | LangChain | LLMs | FAISS | ChromaDB | Docker

- Developed an **AI-powered document question-answering platform** that allows users to upload documents and ask questions using natural language.
- Implemented a **Retrieval-Augmented Generation (RAG)** pipeline combining document processing, text chunking, embeddings, vector similarity search, and LLM-based response generation.
- Used **Sentence Transformers** to generate semantic embeddings and implemented vector retrieval using **FAISS and ChromaDB**.
- Built a **FastAPI backend** for API-based document processing and question answering.
- Developed an interactive **Streamlit interface** for document upload and AI-powered question answering.

2. Plant Disease Detection Using CNN

Python | TensorFlow | Keras | CNN | Jupyter Notebook | Image Classification

- Developed a **Convolutional Neural Network (CNN)** model for automated plant disease classification from leaf images.
- Built the complete deep-learning workflow in **Jupyter Notebook**, including dataset preparation, preprocessing, augmentation, model training, validation, and evaluation.
- Applied image normalization and preprocessing techniques to prepare plant leaf images for deep-learning training.
- Designed a CNN architecture capable of learning visual features associated with different plant diseases.
- Evaluated model performance using validation metrics and analyzed classification results.
- Created a foundation for an AI-based agricultural disease detection application that can assist in early identification of plant diseases.

3. AI-Powered Portfolio Assistant

React | JavaScript | AI/LLM | HTML | CSS | Vercel | Render

- Designed and developed an **AI-powered personal portfolio website** to present technical skills, projects, experience, and AI capabilities.
- Built an interactive frontend using **React** with a modern responsive user interface.
- Integrated an AI assistant concept to provide visitors with an interactive way to explore portfolio information.
- Structured project sections to provide direct access to deployed AI applications and demonstrations.
- Deployed the portfolio as a web application and optimized it as a professional personal branding platform.

4. Wine Quality Prediction using Machine Learning (GitHub)

Python | Pandas | NumPy | Scikit-learn | EDA | Decision Tree | Random Forest | GitHub | Streamlit

- Built a Wine Quality Prediction model using Machine Learning algorithms (Random Forest, Logistic Regression), achieving high classification accuracy on real-world chemical data.
- Performed data preprocessing and feature engineering, including handling missing values, scaling, and outlier analysis to improve model performance and reliability.
- Evaluated models using confusion matrix, precision, and accuracy metrics, and identified Random Forest as the bestperforming model for prediction.

EDUCATION

B.Tech - Computer Science and Engineering 2021 - 2025

Jyothismahthi Institute of Technology and Science, JNTUH | 65%

CERTIFICATIONS

- Datavalley AI Engineer Professional Training Programme (Nov 2025 - Present)
- Python Programming - Sanfoundry Certification
- Python and Data Structures and Algorithms – CodeTantra
- Cloud Computing - Sanfoundry Certification